MYOPIA: Understanding the Limits of Tree-Based Reasoning in Vision-Language Models

Supplementary Material

S1 Scope and Conventions

This supplement provides the complete results referenced in the main paper's **Reproducibility paragraph**:

- Section S2 reports results for all available prompting conditions at $T{=}0$ and $T{=}0.7$, including the tree-with-history ablation.
- Section S3 reports paired-bootstrap intervals and Holm-corrected tests for the tree–zero-shot gap.
- Section S4 evaluates ten paraphrases of the zero-shot prompt, quantifies the resulting spread, and locates the prompt reported in the main paper within that spread.
- Section S5 explains that `RunId` indexes distinct prompt wordings, not repeated calls with the same prompt.
- Sections S6 and S7 report the global-label and numbered-choice probes, respectively.
- Sections S8–S9 document the tree specifications and audit coverage.
- Section S10 describes the released artifacts, fixed image subsets, decoding settings, and limits on exact re-execution.

The following conventions apply to all reported results.

Accuracy denominator. A prediction is correct only when its parsed integer class identifier matches the ground-truth identifier. The denominator includes every row in a condition; unparseable responses and images without a logged traversal are counted as errors. Reported end-to-end accuracy therefore incorporates answer-interface failures as well as classification failures. Interface effects are analyzed separately in the main paper and in Section S7.

Deduplication. Rows in all_sequences_training/ duplicate current GTSRB images in datasets/GTSRB/Training/. All tables use a view that removes a legacy row when a matching current row exists for the same model, temperature, ablation, and run. Without this step, GTSRB denominators increase from 901 to as many as 915 and accuracies shift by up to 0.3 points.

Coverage. A full pass contains 901 GTSRB images, 1,000 MNIST images, 1,000 CIFAR-10 images, and 850 MSCOCO images. This supplement reports every available cell and marks incomplete cells with their n.

S2 Complete Results at Both Temperatures

Table S1 reports six prompting conditions at $T{=}0$ and $T{=}0.7$, extending Table 1 of the main paper with the recorded $T{=}0$ results and the tree-with-history (Tree+H) ablation.

At $T{=}0$, only GTSRB and CIFAR-10 have complete passes. The available MNIST and MSCOCO cells contain 2–860 images and should not be interpreted as dataset-level estimates; they are included with their n for transparency. Accordingly, the main paper discusses $T{=}0$ only for GTSRB and CIFAR-10.

Tree+H likewise has complete coverage only for GTSRB and CIFAR-10. At $T{=}0.7$, it reduces accuracy in five of the six dataset–model comparisons; Qwen-VL on GTSRB is the sole exception. The MNIST and MSCOCO Tree+H cells contain only 2–33 images and are reported as incomplete observations.

Boldface and underlining identify the best and second-best eligible entries in each row. Full-pass conditions and the planned $n{=}100$ CoT probes are eligible; superscripted incomplete cells are excluded.

S3 Significance of the Tree–Zero-Shot Gap

Table S1 shows the tree condition below zero-shot almost everywhere. Table S2 attaches an interval and a corrected p-value to that gap, so the claim rests on more than a point estimate.

The design is paired: for a given dataset, model, and temperature, the zero-shot and tree predictions for one image are stored on the same database row, so the two conditions see identical images. The bootstrap therefore resamples *images*, not predictions, and recomputes both accuracies on each resample; the reported interval is over the difference, which absorbs the correlation between the two conditions rather than treating them as independent samples.

Two limits are worth stating. With 10,000 resamples, the two-sided bootstrap p cannot be resolved below $2/10{,}000$. Eleven comparisons reach this floor, so their corrected values are reported as upper bounds rather than exact quantities. The intervals quantify sampling variation over images at a fixed decoding temperature.

All twelve gaps are negative, and all twelve remain significant after Holm–Bonferroni correction over the family of twelve comparisons (largest adjusted $p = 0.0278$, GTSRB with LLaMA). The single smallest gap, -3.88 points for

Table S1: Accuracy (%) across temperatures and prompting conditions. ZS: zero-shot; +D: class and image descriptions; +H: conversation history across tree nodes; CoT: chain-of-thought. Superscripts give n for incomplete cells, and an em dash indicates no data. CoT uses n=100 class-stratified images per dataset–model pair at T=0.7. Boldface and underlining identify the best and second-best entries among full-pass conditions and the planned CoT probes; superscripted cells are excluded. Full-pass sizes are 901 (GTSRB), 1,000 (MNIST and CIFAR-10), and 850 (MSCOCO).

Dataset	Model	T	ZS	ZS+D	CoT	Tree	Tree+H	Tree+D
GTSRB	GPT-4o	0	**70.92**	63.82	—	58.05	55.83	<u>66.81</u>
		0.7	**71.14**	63.60	63.00	57.71	56.71	<u>65.04</u>
	LLaMA	0	**23.86**	<u>18.42</u>	—	6.33	16.09	15.65
		0.7	**37.07**	23.53	12.00	<u>33.19</u>	12.43	14.87
	Qwen-VL	0	47.95	<u>49.72</u>	—	40.51	36.96	**51.83**
		0.7	47.06	<u>48.95</u>	36.00	24.64	36.96	**52.28**
MNIST	GPT-4o	0	94.87^{234}	88.02^{860}	—	35.47^{234}	46.67^{15}	—
		0.7	<u>91.50</u>	86.40	**94.00**	34.60	75.00^{16}	28.60
	LLaMA	0	33.33^{3}	—	—	33.33^{3}	0.00^{2}	—
		0.7	<u>59.30</u>	40.20	**71.00**	7.70	50.00^{2}	2.60
	Qwen-VL	0	95.17^{414}	—	—	14.25^{414}	—	—
		0.7	**94.40**	<u>89.20</u>	86.00	29.30	7.69^{13}	27.10
CIFAR-10	GPT-4o	0	<u>89.60</u>	**90.00**	—	75.40	73.10	82.70
		0.7	<u>87.90</u>	**88.90**	87.00	76.00	72.80	80.60
	LLaMA	0	<u>43.70</u>	**62.80**	—	17.10	12.30	4.80
		0.7	36.70	**54.00**	<u>38.00</u>	16.40	14.10	6.00
	Qwen-VL	0	**77.60**	<u>71.20</u>	—	56.60	33.00	66.50
		0.7	**77.60**	<u>70.90</u>	63.00	56.60	33.00	66.30
MSCOCO	GPT-4o	0	73.91^{92}	—	—	46.74^{92}	0.00^{7}	—
		0.7	<u>72.82</u>	**80.82**	71.00	46.71	33.33^{33}	44.00
	LLaMA	0	50.00^{2}	—	—	0.00^{2}	0.00^{4}	—
		0.7	**23.76**	<u>21.06</u>	18.00	3.65	33.33^{6}	0.35
	Qwen-VL	0	76.47^{221}	—	—	40.27^{221}	—	—
		0.7	<u>74.82</u>	**78.35**	59.00	39.65	—	44.47

LLaMA on GTSRB, is also the only one whose interval approaches zero ($[-7.33, -0.44]$); every other interval excludes zero by a wide margin.

S4 Ten-Prompt Zero-Shot Sweep

Section 5.1 of the main paper evaluates ten zero-shot prompts at T=0.7 and reports one of them in Table 1. This section gives the full sweep behind that choice. Ten paraphrases of the zero-shot instruction were written per dataset; they differ in register and verbosity but request the same output, a single class ID. The prompt index is stored as `RunId`, and prompt 0 is the one reported in main-paper Table 1.

Table S3 and Figure S1 give per-prompt accuracy for the six dataset–model pairs swept at full scale. Five pairs contain ten full-coverage prompts; GPT-4o on GTSRB contains eight, because P2 and P8 stopped at n=78 and n=108 respectively. Summary statistics exclude those two cells, and the partial accuracies are not comparable to the rest of the row.

Prompt sensitivity is a first-order effect. Across-prompt standard deviations range from 2.81 points (CIFAR-10, GPT-4o) to 16.97 points (GTSRB, Qwen-VL), and the max–min spread reaches 41.5 points on GTSRB for Qwen-VL and 31.1 points for LLaMA. Prompt choice is therefore comparable in magnitude to the differences between models, and any single zero-shot number should be read as one draw from this spread. Part of the spread is discontinuous rather than gradual: Qwen-VL falls to 5.55% on one GTSRB paraphrase and to 50.10% on one CIFAR-10 paraphrase, drops more consistent with format non-compliance than with a loss of recognition ability.

Where prompt 0 sits in the distribution. On CIFAR-10, prompt 0 is close to the sweep mean for GPT-4o (87.90% versus 87.08%), below it for LLaMA (36.70% versus 42.13%), and above it for Qwen-VL (77.60% versus 71.01%). On GTSRB it is the highest full-coverage prompt for all three models: 71.14%, 37.07%, and 47.06%, against sweep means of 59.67%, 16.57%, and 35.84%. The GTSRB zero-shot column of main-paper Table 1 is therefore drawn from the favorable end of its prompt distribution, while the CIFAR-10 column is representative.

Effect on the Tree–zero-shot comparison. Tree traversal was evaluated once, with a fixed node prompt that does not vary with `RunId` (Section S5), so the sweep bears on only one side of the comparison. On GTSRB, mean zero-shot accuracy stays above tree accuracy for GPT-4o (59.67% versus

Table S2: Paired bootstrap over images for the tree-minus-zero-shot accuracy gap at $T=0.7$ (no descriptions, no history, run 0). Images are resampled with replacement (10,000 resamples; seed 42), and both accuracies are recomputed on each resample. Δ is Tree$-$ZS in percentage points; the interval is the 2.5th–97.5th percentile range of the resampled Δ. p_{Holm} is the two-sided bootstrap p after Holm–Bonferroni correction across the twelve dataset–model comparisons. Values preceded by $<$ are upper bounds imposed by bootstrap resolution.

Dataset	Model	n	ZS	Tree	Δ	95% CI	p_{Holm}
GTSRB	GPT-4o	901	71.14	57.71	-13.43	$[-16.43, -10.54]$	<0.0024
GTSRB	LLaMA	901	37.07	33.19	-3.88	$[-7.33, -0.44]$	0.0278
GTSRB	Qwen-VL	901	47.06	24.64	-22.42	$[-25.64, -19.20]$	<0.0024
MNIST	GPT-4o	1000	91.50	34.60	-56.90	$[-60.00, -53.70]$	<0.0024
MNIST	LLaMA	1000	59.30	7.70	-51.60	$[-55.00, -48.10]$	<0.0024
MNIST	Qwen-VL	1000	94.40	29.30	-65.10	$[-68.10, -62.10]$	<0.0024
CIFAR-10	GPT-4o	1000	87.90	76.00	-11.90	$[-14.50, -9.40]$	<0.0024
CIFAR-10	LLaMA	1000	36.70	16.40	-20.30	$[-23.80, -16.80]$	<0.0024
CIFAR-10	Qwen-VL	1000	77.60	56.60	-21.00	$[-24.10, -17.90]$	<0.0024
MSCOCO	GPT-4o	850	72.82	46.71	-26.12	$[-29.88, -22.47]$	<0.0024
MSCOCO	LLaMA	850	23.76	3.65	-20.12	$[-23.06, -17.18]$	<0.0024
MSCOCO	Qwen-VL	850	74.82	39.65	-35.18	$[-38.82, -31.41]$	<0.0024

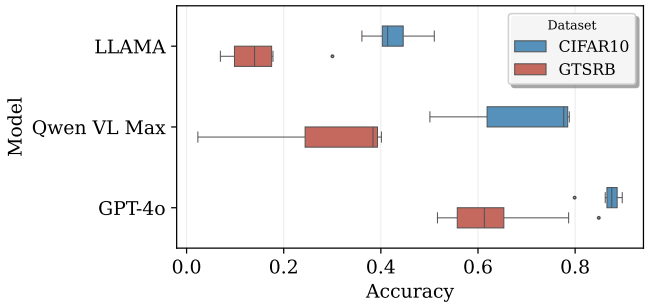


Figure S1: Prompt sensitivity of zero-shot accuracy at $T=0.7$. Dots show full-coverage prompt results, diamonds mark prompt 0 (reported in main-paper Table 1), black ticks show sweep means, and horizontal lines span the observed range. Red denotes GTSRB and blue denotes CIFAR-10. On GTSRB, prompt 0 is the highest full-coverage prompt for all three models; the GPT-4o/GTSRB row spans eight prompts rather than ten.

57.71%) and Qwen-VL (35.84% versus 24.64%), but falls below it for LLaMA (16.57% versus 33.19%); several individual prompts also reverse the GPT-4o and Qwen-VL gaps. On CIFAR-10 the zero-shot mean stays above tree accuracy for all three models. The twelve negative gaps in Table S2 thus describe the prompt 0 comparison specifically. The direction of the result is robust on CIFAR-10 and for the aggregate, but on GTSRB its magnitude, and for LLaMA its sign, depend on zero-shot prompt wording.

S5 Scope of Repeated Measurement

The repeated-measurement field in the released databases does not index independent stochastic replications. `RunId` selects the zero-shot prompt paraphrase, and all ten indices correspond to distinct prompt texts. The variation reported in Section S4 therefore measures sensitivity to wording, not variation across repeated calls with an unchanged prompt.

Tree traversal was executed under `RunId` 0 alone. Its prompt is built from the current node's question and answer set and does not vary with the zero-shot prompt index, so the database contains exactly one tree traversal per dataset–model–temperature–ablation cell. Reported tree accuracies are single-run measurements. Replicated tree runs remain an important robustness check.

Where a full-scale sweep is unavailable, Table S4 reports the available prompt checks on MNIST and MSCOCO, restricted to the images shared by all prompts so that the spread reflects wording rather than differing image subsets. Three of the four checks have SD $\leq$ 2.9 points; GPT-4o on MSCOCO has SD 4.79 points over a 65.0–75.0% range. Each check uses 20 images, so these estimates are imprecise and do not replace a full-scale sweep.

S6 Global-Label Probe

Tree traversal shows the model only the current node's answer options, whereas zero-shot prompting shows the complete label set. If restricted label access caused most of the Tree–zero-shot gap, appending all labels to each node prompt should recover a substantial part of it. We test this hypothesis with GPT-4o on matched, class-stratified subsamples of 20–50 images per dataset. Each image is evaluated once with the local options and once with the global label list appended.

Table S5 reports both conditions on their shared images. In dataset order (GTSRB, MNIST, CIFAR-10, MSCOCO), the global list changes node accuracy by $+1.0$, -5.3, -1.4, and $+0.1$ points; image accuracy changes by $+4$, 0, -5, and $+4$ points. The unresolved rate also increases on three of the four datasets, consistent with the longer prompt making format compliance more difficult in some cases.

The probe therefore provides no evidence that label access

Table S3: Zero-shot accuracy (%) for each of the ten prompt paraphrases at T=0.7, on the six dataset–model pairs swept at full scale. P0, underlined, is the prompt reported in main-paper Table 1. Em dashes mark runs that did not reach full coverage and are excluded from the summary statistics. Mean, SD, and Δ are computed over the available prompts, with $\Delta = \max - \min$.

Dataset	Model	P0	P1	P2	P3	P4	P5	P6	P7	P8	P9	Mean	SD	Δ
GTSRB	GPT-4o	71.14	63.71	—	66.04	45.84	56.16	67.70	49.94	—	56.83	59.67	8.94	25.31
	LLaMA	37.07	23.20	15.54	20.75	11.10	17.09	18.87	8.10	5.99	7.99	16.57	9.28	31.08
	Qwen-VL	47.06	46.95	46.06	45.84	5.55	45.84	46.06	11.88	17.31	45.84	35.84	16.97	41.51
CIFAR-10	GPT-4o	87.90	86.20	88.70	86.50	89.70	89.50	86.70	87.20	79.90	88.50	87.08	2.81	9.80
	LLaMA	36.70	51.00	43.10	36.10	41.20	41.60	40.90	45.10	40.10	45.50	42.13	4.38	14.90
	Qwen-VL	77.60	78.40	76.50	50.10	77.70	57.00	78.50	56.90	78.60	78.80	71.01	11.45	28.70

Table S4: Small-scale zero-shot prompt checks on the datasets without a full-scale sweep. Each row compares four prompts on the same matched image subset, so the spread is not confounded by which images each prompt saw. Mean, SD, and Δ are percentages, with $\Delta = \max - \min$.

Dataset	Model	n	Prompts	Mean	SD	Δ
MNIST	GPT-4o	20	4	93.75	2.50	5.00
	Qwen-VL	20	4	97.50	2.89	5.00
MSCOCO	GPT-4o	20	4	68.75	4.79	10.00
	Qwen-VL	20	4	68.75	2.50	5.00

Table S5: Global-label probe with GPT-4o on images shared by both conditions. Local uses only the current node's answer options; Global also appends the dataset's full class list. Node accuracy and unresolved rate are computed over visited node decisions, and image accuracy is end-to-end traversal accuracy. All values are percentages.

		Node acc.		Unresolved		Image acc.	
Dataset	n_{img}	Local	Global	Local	Global	Local	Global
GTSRB	50	89.1	90.1	2.1	3.1	58.0	62.0
MNIST	20	71.1	65.8	0.0	0.0	35.0	35.0
CIFAR-10	20	88.9	87.5	1.6	7.8	65.0	60.0
MSCOCO	50	91.2	91.3	0.3	3.8	46.0	50.0

Table S6: Numbered-choice probe with GPT-4o (n=20 images per dataset). Free uses strict extraction from a free-form response; Num. requests only an option number. Node accuracy and unresolved rate use each condition's visited nodes; image accuracy is end-to-end. All values are percentages.

		Node acc.		Unresolved		Image acc.	
Dataset	n_{img}	Free	Num.	Free	Num.	Free	Num.
GTSRB	20	89.0	90.7	1.4	0.0	60.0	65.0
MNIST	20	68.3	71.4	2.4	0.0	35.0	30.0
CIFAR-10	20	92.3	92.2	1.5	0.0	75.0	75.0
MSCOCO	20	88.2	86.7	0.0	0.0	35.0	30.0

is the primary cause of the gap: it produces no large or consistent recovery toward the full-run zero-shot results. Because the matched samples are small, however, the probe cannot resolve effects of only a few points and should not be read as evidence of exactly zero effect.

Table S7: Numbered-choice probe on nodes reached at the same image and depth in both conditions. Δ is numbered minus free-form node accuracy, in percentage points.

Dataset	Paired nodes	Free	Num.	Δ
GTSRB	73	89.0	90.4	+1.4
MNIST	36	69.4	77.8	+8.3
CIFAR-10	63	92.1	92.1	+0.0
MSCOCO	105	90.5	86.7	-3.8

S7 Numbered-Choice Probe

Because the strict extractor marks a response unresolved whenever it fails to match a predefined answer, part of the tree deficit could be an answer-interface artifact rather than a reasoning failure. This probe controls for that possibility by asking each node twice on the same 20 images per dataset: once with the free-form prompt and strict regular-expression extraction, and once with a numbered-option prompt that requests only an option number.

Table S6 reports both conditions. The numbered format eliminates unresolved responses (0.0% on all four datasets), while free-form responses are already nearly fully parseable for GPT-4o (0.0–2.4% unresolved). The corresponding node-accuracy changes are $+1.7$, $+3.1$, -0.1, and -1.5 points for GTSRB, MNIST, CIFAR-10, and MSCOCO, respectively. Removing unresolved responses therefore does not produce a consistent accuracy improvement.

A caveat on matching. Each traversal terminates at its first incorrect or unresolved decision, so the two conditions do not visit the same number of nodes even on the same image: a condition that fails earlier accumulates fewer, shallower, and therefore easier nodes. The node accuracies in Table S6 are computed over each condition's own node set, which is what the main paper reports. Table S7 instead restricts the analysis to nodes reached at the same depth on the same image in both conditions. Matching in this stricter sense increases the MNIST gain from $+3.1$ to $+8.3$ points and deepens the MSCOCO loss from -1.5 to -3.8 points, while GTSRB and CIFAR-10 are essentially unchanged.

A caveat on power. The probe uses 20 images per dataset, yielding 36–110 node decisions per condition. None of

the four exploratory differences is distinguishable from zero (Fisher exact, all $p > 0.75$), and the image-level direction reverses on two datasets. These tests also treat node decisions as independent and do not model their nesting within images. The probe therefore rules out only a large interface effect for GPT-4o; it does not establish equivalence between the formats. Notably, the free-form unresolved rates are well below GPT-4o's 9.9% rate in the full GTSRB run, indicating that its interface failures are concentrated in particular conditions rather than inherent to free-form prompting.

S8 Tree Specifications

Table S8 summarizes the four evaluated trees. Every internal node has at least two children, every leaf maps to exactly one class, and no question repeats along a root-to-leaf path. In the branching column, for example, 16×2 denotes 16 binary internal nodes.

For an internal node v, let p_{vj} be the fraction of descendant leaves assigned to child j. We define split entropy as $H_v = -\sum_j p_{vj} \log_2 p_{vj}$. Higher values indicate a more even division of the remaining classes. The values are not normalized by the maximum entropy for the node's branching factor.

The trees differ substantially in shape. MNIST and CIFAR-10 are shallow (maximum depth 3) with 8–9 internal nodes. MSCOCO is a binary tree with 89 internal nodes and maximum depth 10. GTSRB is the most irregular: it combines 16 binary nodes with two ternary, two five-way, and two eight-way nodes and reaches depth 14. An independent LLM judge flagged the eight-way GTSRB arrow-direction question as a possible question-design concern; it is also the highest-entropy split in the four trees (3.00 bits).

Five MSCOCO internal nodes were never reached by a primary-condition traversal. Because they have no measured traversal accuracy, they are excluded from the audit, leaving 123 of 128 internal nodes with operational estimates.

S9 Tree-Quality Audit Coverage

Table S9 shows how the 347 retained node–model observations are distributed across datasets and models. The audit covers 123 distinct internal nodes, yielding 369 possible node–model combinations. A combination is retained only when both conditional in-traversal accuracy and a grounding-gap estimate are available; 22 combinations are therefore excluded.

Coverage is uneven. MSCOCO contributes 237 of the 347 observations (68%) from 84 nodes, whereas MNIST contributes 22 observations from 8 nodes. Any pooled association is therefore weighted toward MSCOCO, the only deep binary tree in the set. The audit's conclusions are accordingly limited to these four evaluated trees rather than presented as general properties of hierarchical classifiers.

Per-model coverage is 123 observations for GPT-4o, 122 for Qwen-VL Max, and 102 for LLaMA-3.2. Because traversal stops after the first failure, node accuracy is conditional on reach: deeper nodes are scored only on traversals that completed all earlier decisions correctly. It should therefore be read as an operational quantity, not as an intrinsic measure of node answerability.

S10 Reproducibility Details

This section records what a reader needs to regenerate the reported results and which parts of the evaluation cannot be reproduced exactly.

Artifacts in this supplement. The accompanying archive contains the four evaluated decision trees in both human-readable and machine-readable forms; the complete prompt set, including zero-shot paraphrases, per-class descriptions, and node-question templates; the generator–evaluator tree-construction pipeline; the evaluation harness; and the analysis scripts that produce every table in this document. It excludes API credentials, raw provider traces, and the datasets themselves.

Sampling. Each dataset is evaluated on a fixed image subset listed by identifier in the archive: 901 GTSRB, 1,000 MNIST, 1,000 CIFAR-10, and 850 MSCOCO images. The MSCOCO subset is restricted to images with exactly one category label, ensuring that each image has a single ground-truth class. The identifier lists constitute the sampling record; the subsets are not redrawn at run time.

Decoding. All primary conditions use temperature 0 or 0.7, one image per request, no few-shot examples, and the endpoint or deployment defaults for other decoding parameters. Tree traversal issues one request per visited node. The tree-with-history ablation additionally carries previous node questions and answers forward in the conversation.

Determinism and its limits. The analysis layer is deterministic: the bootstrap uses seed 42 and 10,000 resamples, and every table is recomputed from the stored databases rather than from cached summaries. Model inference is not guaranteed to be bit-reproducible, even at $T{=}0$. GPT-4o and Qwen-VL Max are accessed through hosted services whose implementations or checkpoints may change. LLaMA-3.2 11B Vision Instruct is self-hosted from fixed official weights at FP16 precision, but hardware and software nondeterminism can still affect its outputs. A new evaluation run may preserve the qualitative patterns without reproducing the exact percentages.

Recomputation path. The stored databases are the authoritative source for every table reported here. Given these files, the archived analysis scripts regenerate all tables and the figure without model access or API credentials. Reproducing the databases themselves requires access to the hosted services and the self-hosted LLaMA deployment, with the associated API and compute costs.

Table S8: Specifications of the four evaluated trees. Int. is the number of internal nodes; Depth is the maximum root-to-leaf depth. Branching gives internal-node outdegrees and their multiplicities. Split entropy is reported in unnormalized bits. Unreached counts internal nodes that no primary-condition traversal visited.

Dataset	Int.	Leaves	Depth	Branching	$\bar{H}$	H range	Unreached
GTSRB	22	43	14	16×2, 2×3, 2×5, 2×8	1.02	0.35–3.00	0
MNIST	8	10	3	7×2, 1×3	1.01	0.92–1.30	0
CIFAR-10	9	10	3	9×2	0.96	0.81–1.00	0
MSCOCO	89	90	10	89×2	0.93	0.54–1.00	5
Total	**128**	**153**					**5**

Table S9: Tree-quality audit coverage. Dataset rows report distinct retained nodes and node–model pairs; model rows report retained pairs. Each pair requires both conditional in-traversal accuracy and a grounding gap.

Dataset	Nodes	Node–model pairs
By dataset		
GTSRB	22	61
MNIST	8	22
CIFAR-10	9	27
MSCOCO	84	237
By model		
GPT-4o		123
LLaMA		102
Qwen-VL		122
Total	**123**	**347**

Scoring requirements. Exact recomputation requires the GTSRB deduplication described in Section S1; otherwise, denominators increase from 901 to as many as 915. Accuracy must also count unparseable answers and unlogged traversals as errors. Restricting evaluation to parseable responses raises the tree results and makes them incomparable with the values reported here.